

AtmoBench Overview

1 Multi-Region Air Quality Dataset

Data Sources



Dataset Statistics

Stations	2,812
Pollutants	6
Time Series	14,139

Data Processing

Cleaning

Harmonization

Missing Value Handling

2 Forecasting Benchmark Framework

Data Split

Training
(1 Year)

Testing
(2 Years)

Evaluation Settings

Zero-Shot

Evaluate directly

Fine-Tuning

Adapt on 1-year data

Models Compared

Time Series Foundation Models

- VisionTS++
- TiRex
- TimesFM-2.5
- Moirai-2
- Chronos-2
- Chronos-Bolt
- Sundial
- TimesFM-2.0
- Moirai-1
- TimesFM-1.0
- Kairos

ML/DL Models

- PatchTST
- DLinear
- DeepAR
- LightGBM

Statistical Baseline Models

- AutoETS
- Seasonal Naive

3 Performance Evaluation

Evaluation Metrics

MASE

CRPS

MAE

RMSE

sMAPE

Comprehensive Comparisons

Overall Ranking

Pollutant-wise Comparison

Region-wise Comparison

Station-wise Comparison

Outputs

Leaderboard

Model Ranking

Performance Curves

Error Trends

4 Error Analysis & Insights

Bias Analysis

Pollutant Bias

Country Bias

Station Bias

Urban vs. Rural Bias

Error Analysis

Geographic (Map)

Seasonal

Forecast Horizon

Error Distribution

Key Insights

- Most difficult pollutants
- Most challenging regions/stations
- TSFM strengths
- TSFM weaknesses
- Guidance for future model development